

7. Analysis of Convertible Bonds

1. In the October 26, 1992, prospectus summary of the Staples 5% convertible subordinated debentures due 1999, the offering stated: “Convertible into Common Stock at a conversion price of \$45 per share . . .” If the par value is \$1,000, what is the conversion ratio?

Let us rearrange the equation for conversion price to solve for the conversion ratio. We have:

$$\text{conversion price} = \frac{\text{par value}}{\text{conversion ratio}} \rightarrow \text{conversion ratio} = \frac{\text{par value}}{\text{conversion price}}.$$

Inserting in our given values, we have:

$$\text{conversion ratio} = \frac{\text{par value}}{\text{conversion price}} = \frac{\$1,000}{45} = 22.22.$$

2. Consider a convertible bond as follows:

par value = \$1,000; coupon rate = 9.5%
market price of convertible bond = \$1,000
conversion ratio = 37.383
estimated straight value of bond = \$510
yield to maturity of straight bond = 18.7%

Assume that the price of the common stock is \$23 and that the dividend per share is \$0.75 per year.

Calculate each of the following (1) conversion value, (2) market conversion price, (3), conversion premium per share, (5) conversion premium ratio, (5) premium over straight value, (6) favorable income differential per share, and (7) premium payback period.

(1) conversion value = market price of common stock $\times$ conversion ratio = $\$23 \times 37.383 = \859.809 or about **\$859.81**.

(2) market conversion price = $\frac{\text{market price of convertible bond}}{\text{conversion ratio}} = \frac{\$1,000}{37.383} =$

\$26.750127 or about **\$26.75**.

(3) market conversion premium per share = market conversion price – current market price = \$26.7501127 – \$23 = \$3.750127 or about **\$3.75**.

(4) market conversion premium ratio = $\frac{\text{conversion premium per share}}{\text{market price of common stock}} =$
 $\frac{\$3.750127}{\$23} = 0.16305$ or about **16.3%**.

(5) premium over straight value = $\left(\frac{\text{market price of the convertible bond}}{\text{straight value}} \right) - 1 =$
 $\frac{\$1,000}{\$510} - 1 = 1.96078 - 1 = 0.96078$ or about **96.08%**.

(6) Noting that the conversion ratio = $\frac{\text{par value}}{\text{conversion price}} = \frac{\$1,000}{26.750127} =$

37.383, we can insert all of our given values into our favorable income differential per share to get:

$$\frac{(\text{coupon rate} \times \text{par value}) - (\text{conversion ratio} \times \text{dividend per share})}{\text{conversion ratio}} =$$

$$\frac{\$95 - (37.383 \times \$0.75)}{37.383} = \$1.7912621 \text{ or about } \mathbf{\$1.79}.$$

(7) premium payback period = $\frac{\text{market conversion premium per share}}{\text{favorable income differential per share}} =$
 $\frac{\$3.750127}{\$1.7912621} = 2.09357$ or about **2.09**.

3. A Merrill Lynch note structure called a liquid yield option note (LYON) is a zero-coupon instrument that is convertible into the common stock of the issuer. The conversion ratio is fixed for the entire life of the note. If investors wish to convert to the shares of the issuer, they must exchange the LYON for the stock. As a result, the market conversion price increases over time. Why?

The formula for market conversion price is:

$$\text{Market conversion price} = (\text{market price of bond}) / (\text{conversion ratio}).$$

For a zero-coupon bond, the market price of the bond increases over time since interest is not paid in cash but increases the value of the bond. Thus, from the above formula for the market conversion price, we can see that as the market price of the bond increases that the market conversion price must increase proportionally since the conversion ratio is constant.

4. What is a cash flow arbitrage strategy involving convertible bonds?

The cash flow from a convertible bond differs from the cash flow from a stock. The idea behind a **cash flow arbitrage strategy** is to create equivalent positions in the convertible bond and underlying stock so that any additional cash flow available from the convertible bond can be captured while eliminating or mitigating any risks. More specifically, the convertible bond is purchased with funds obtained from shorting the underlying stock.

In convertible bond arbitrage trades, typically a long position is established in the convertible bond, and simultaneously, a short position is established in the underlying stock. The number of shares shorted is such that any change in the value of the convertible bond will be equal to the change in the position of the stocks shorted. This is determined by the delta of the convertible bond, and the resulting position is said to be market neutral.